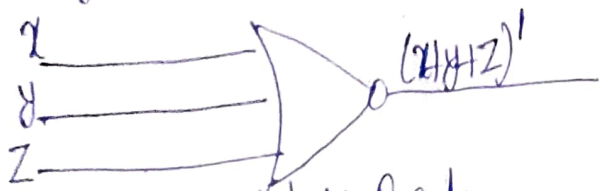
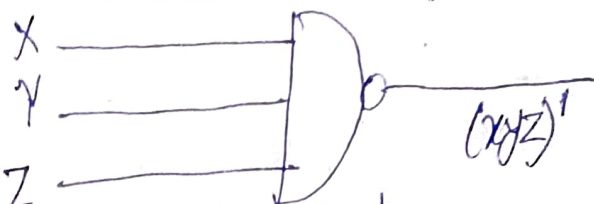


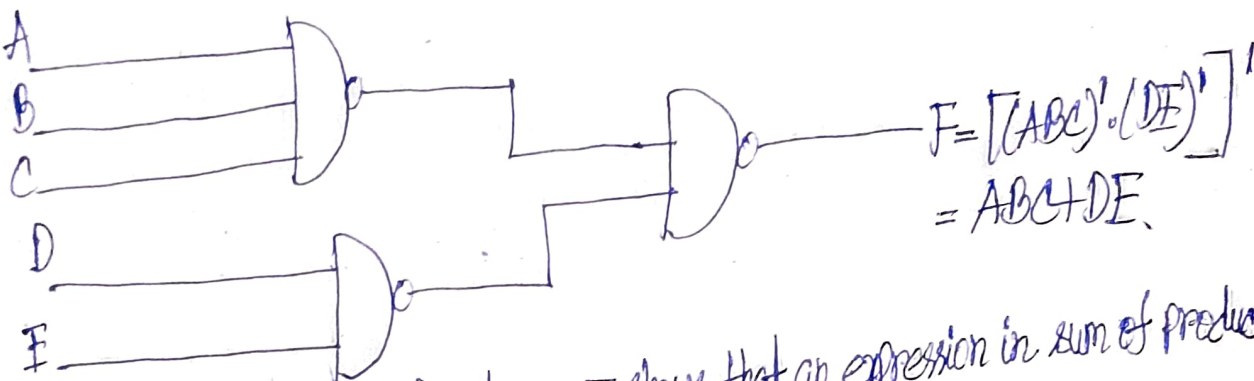
Figure 2.7 Multiple-input and Cascaded NOR and NAND gate



(a) Three-input NOR gate

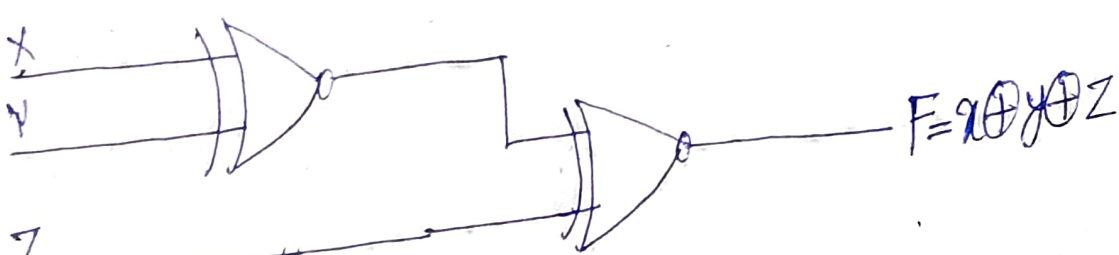


(b) Three-input NAND gate



(c) Cascaded NAND gate $\rightarrow$ shows that an expression in sum of products can be implemented with NAND gates.

Figure 2.8 Three input exclusive-OR gate



(a) Using two-input gates

(b) A three-input gate

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

The exclusive-OR and equivalence gates are both commutative and associative and can be extended to more than two inputs.